

Sheng-Kai (Kyle) Wen

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Experience

PuckAI, Live AI Platform

May 2025 - Present

AI Engineer, Independent (via UMass CICS CICStep)

thepuckai.com

- **Built and operated a production Retrieval-Augmented Generation (RAG) chatbot** (FastAPI backend, Next.js / TypeScript frontend) over **7,643** grounded reports, with **hybrid dense (pgvector HNSW) and lexical retrieval** fused by weighted Reciprocal Rank Fusion, **Voyage AI** embeddings, and a cross-encoder reranker narrowing 20 candidates to 5.
- **Took scope-limited query relevance from NDCG@10 0.376/0.307 to 1.000** on a **77-query benchmark** by engineering a **deterministic alias-index router** feeding a SQL pre-filter, after diagnosing that embeddings cannot encode current team membership, with name and archetype queries unchanged to three decimals.
- **Shipped 7,643 grounded reports** (7,643 validated, 0 quarantined) by **constraining the LLM to pipeline-computed numbers only** behind a deterministic validator (numeric trace, proper-noun trace, banned vocabulary) in a mandatory self-check loop.
- **Built a retrieval evaluation harness (NDCG@10, MRR, Recall@K, P@K)** and used it to raise end-to-end relevance **NDCG@10 0.488 to 0.682** across a five-configuration ladder, catching multiple silent retrieval-correctness regressions before they shipped.
- **Tuned fusion to a 2:1 dense-to-lexical ratio** after measuring equal weighting underperformed *both* legs alone, and **capped LLM spend** behind per-user quotas plus a post-retrieval relevance gate that refuses off-topic questions before any completion.

Honda Research Institute (HRI) & 99P Labs

Feb 2026 - May 2026

Graduate Research Extern, Hindsight team (Advised by Prof. Andrew McCallum)

- **Architected Hindsight, a knowledge-graph retrieval layer** fusing **BM25** with agentic, **intent-conditioned graph traversal** (an LLM labels each question leading, lagging, or exploratory to steer edge direction) over open models (Qwen3-8B/32B) with **AST-based indexing**.
- **Built the evaluation infrastructure and metrics** (Fact Recall, Acc@k, LLM-judged answer quality) and **ran ablations across model scales**, reaching file-level **Acc@1 0.535** vs BM25 0.454 (+8.1 pts; +**12.9 pts** with no location hint) over **DeepCodeBench** (1,144 QA pairs) and **LocBench** (560 tasks); **co-authored the 14-page research report**.
- **Traced failing tests to the source locations responsible** by engineering a **LocAgent-based test-linkage layer**, grounding model reasoning in verifiable evidence rather than similarity.

Fan2see, Taipei, Taiwan

Apr 2021 - Apr 2022

Co-Founder & Full-Stack Engineer

- **Shipped real systems under load** by **building a full-stack PERN product** (React, Node.js, PostgreSQL) to hundreds of active users with end-to-end ownership, driving **30% month-over-month growth**.
- **Reduced media-processing latency 25% and improved scalability 40%** by architecting an asynchronous **AWS Lambda/S3 pipeline** for high-volume media workloads.

Applied AI Projects

MAIN-RAG: Multi-Agent Filtering for RAG (UMass Information Retrieval)

Sep 2025 - Dec 2025

- **Reimplemented the Predictor-Judge-Generator pipeline from scratch** with a judge scoring each (query, document, answer) tuple by **logit margin**, then **replaced the expensive LLM judge with rule-based thresholds** (Otsu's method, quantile, mean/std) matching adaptive-threshold quality across **four datasets**.

Timestamp-Aligned Summarization from ASR Transcripts (UMass Advanced NLP)

Sep 2025 - Dec 2025

- **Reached ROUGE-L 59.83 with BERTScore F1 91.44 at 80.43% segment-alignment accuracy** on MeetingBank by **fine-tuning BART** for segment titles with a **BERT-based self-retrieval aligner**.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C/C++

Generative AI & LLM: Retrieval-Augmented Generation (RAG), prompt engineering, grounded generation & deterministic output validation, LLM integration (Claude, GPT-4o, Gemini, Qwen3, Llama-3.1), fine-tuning (BART/BERT), LLM-as-judge, LLM cost control (quotas & relevance gating)

Retrieval & Evaluation: vector search (pgvector, HNSW, FAISS), embeddings & reranking (Voyage AI, Sentence-BERT), hybrid dense + lexical retrieval, Reciprocal Rank Fusion, BM25, DPR, Postgres full-text search, HNSW index tuning, knowledge graphs, AST indexing, NDCG / MRR / Recall@K / Acc@k / ROUGE / BERTScore

Backend, Frontend & Cloud: FastAPI, REST APIs, PostgreSQL, pgvector, Supabase, Next.js (App Router) / React / TypeScript, Docker, CI/CD, AWS (Lambda, S3), Git, pytest, XGBoost, PyTorch

Education

University of Massachusetts Amherst, M.S. Computer Science, GPA 3.83/4.00

Sep 2024 - May 2026

Coursework: Machine Learning, Natural Language Processing (NLP), Information Retrieval, Artificial Intelligence

National Tsing Hua University, M.S. Physics

Sep 2020 - Jan 2023

National Dong Hwa University, B.S. Electrical Engineering, Minor in Computer Science

Sep 2016 - Jun 2020